ETSI TS 128 311 V16.0.0 (2020-11)



5G;
LTE;
Management and orchestration;
Network policy management for mobile networks based on Network Function Virtualization (NFV) scenarios
(3GPP TS 28.311 version 16.0.0 Release 16)



Reference
DTS/TSGS-0528311vg00

Keywords
5G,LTE

ETSI

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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

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The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document contains the architecture, requirements, use cases, procedures and definitions of interfaces for policy management for 4G networks.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] ETSI GR NFV-IFA 023 (V3.1.1): "Network Function Virtualisation (NFV); Management and Orchestration; Report on Policy Management in MANO; Release 3".

[3] 3GPP TS 32.312: "Telecommunication management; Generic Integration Reference Point (IRP) management: Information Service (IS)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

EM	Element Management
IRP	Integration Reference Point
NE	Network Element
NFVI	Network Functions Virtualization Infrastructure
NFVO	Network Functions Virtualization Orchestrator
NM	Network Management
PAF	Policy Administration Function
PF	Policy Function
VIM	Virtualized Infrastructure Manager
VNF	Virtualized Network Function
VNFM	VNF Manager

4 Overview

NFV policy management system may have policy conflict problem, which is a newly created policy may be conflict with an existing policy.

There are two potential policy conflict detection approaches, one is policy conflict detection in the Policy Function (PF), which makes the policy execution decisions, and the other one is policy conflict detection in the Policy Administration Function (PAF), which defines the network policy. There are multiple scenarios for policy conflict. For example, if the events are the same and the same condition are triggered (e.g. scaling policy on a VNF), but two policies have different actions, the conflict should be detected and resolved. It is not easy for one PF to resolve all of scenarios of the policy conflict. Hence, it is reasonable to enable PAF to resolve policy conflict, while the conflict detected by PAF or PF based on different specific scenarios, and the present document try to resolve the problem.

5 Policy management architecture

In order to delegate the support of policy control for virtualized NFs such as automatic scale in/out under certain condition , 3GPP management system would use the ETSI NFV defined related policy features/services. Consequently, two important logical functions are needed for the definition of the 3GPP policy management architecture.

1) Policy Administration Function (PAF) provides the services as follows:

   - Allows the operator to define and administrate the network policy

2) Policy Function (PF) makes the decision for policy execution and provides the services as follows:

   - Activating/deactivating the network policy
   - Event subscription/notification of the network policy execution
   - Providing relevant data and parameters during the process of policy execution (e.g. execution time consuming)

The 3GPP global policy management system would affect the behavior of all 3GPP defined nodes and would delegate the support of policy for virtualized NFs in ETSI MANO systems [2]. For NFV related policy management, 3GPP mainly focus on NM acting as PAF and EM acted as PF, and the interface between the NM and EM can be updated to support policy related operations. Hence, 3GPP policy management architecture under NFV scenario is as follows.

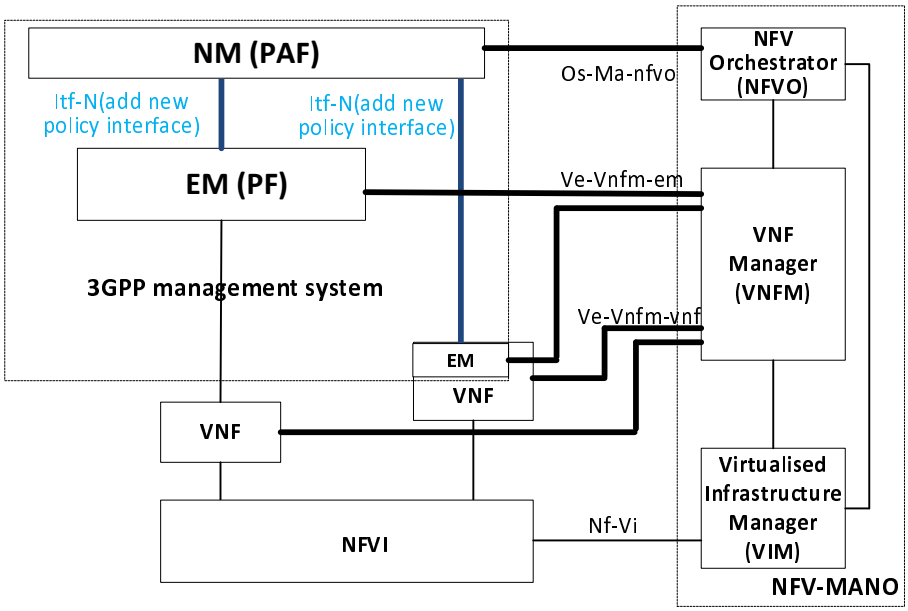


Figure 5-1: NFV policy management architecture in reference point presentation

NFV related policy interfaces between 3GPP management system and MANO such as Os-Ma-nfvo and Ve-Vnfm-em have been defined by ETSI.

6 Business level requirements

6.1 Requirements

REQ-POM_PO-CON-1 3GPP management system should be able to support the capability about the network policy.

6.2 Actor roles

See detailed actors and roles for each use case in clause 6.4.

6.3 Telecommunication resources

See detailed telecommunication resources for each use case in clause 6.4.

6.4 High-level use cases

6.4.1 Deploy the network function

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	According to the network deployment policy, 3GPP system use the policy to deploy the corresponding network.	
Actors and Roles	NM create/delete/update/query/activate/deactivate the network deployment policy, EM execute the NM's policy operation request, and notifies NM policy conflict, if exist.	
Telecom resources	3GPP management system (EM, NM)	
Assumptions	3GPP management system want to deploy a network	
Pre conditions	The operator designs the network deployment requirement and need deploy the network based on the requirement.	
Begins when	There is a new network policy about the network deployment.	
Step 1 (M)	The operator sets up the network deployment policy through NM.	
Step 2 (M)	The NM activates the policy and notifies related EM.	
Step 3 (M)	The EM follows the policy to design the network then the network is instantiated by the management system.	
Ends when	The EM informs the NM that the policy has been executed.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The network is correctly configured and normally running	
Traceability	REQ-POM_PO-CON-1	

7 Specification level requirements

7.1 Requirements

REQ-POM-FUN-01

The IRPAgent shall provide the capability to allow the IRPManager to create the network policy.

REQ-POM-FUN-02

The IRPAgent shall provide the capability to allow the IRPManager to delete the network policy.

REQ-POM-FUN-03

The IRPAgent shall provide the capability to allow the IRPManager to update the network policy.

REQ-POM-FUN-04

The IRPAgent shall provide the capability to allow the IRPManager to query the network policy.

REQ-POM-FUN-05

The IRPAgent shall provide the capability to allow the IRPManager to activate the network policy.

REQ-POM-FUN-06

The IRPAgent shall provide the capability to allow the IRPManager to deactivate the network policy.

REQ-POM-FUN-07

The IRPAgent should notify the IRPManager when there is a network policy conflict.

7.2 Actor roles

See detailed actors and roles for each use case in clause 7.4.

7.3 Telecommunications resources

See detailed telecommunication resources for each use case in clause 7.4.

7.4 Use cases

7.4.1 Create a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To create a policy such as deploying the SBC from IMS and P-GW from EPC in the same Data Centre.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running.	
Begins when	The IRPManager decides to create a policy.	
Step 1 (M)	The IPRManager makes a decision to create a network policy according to the network requirements, such as deploying the SBC.	
Step 2 (M)	After the completion of the policy creation process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy is created and stored in the IRPAgent.	
Traceability	REQ-POM-FUN -01	

7.4.2 Delete a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To delete a policy such as deploying the SBC from IMS and P-GW from EPC in the same Data Center.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is a network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to delete a policy.	
Step 1 (M)	The IPRManager makes a decision to delete the existing policy.	
Step 2 (M)	After the completion of the policy deletion process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy is deleted in the IRPAgent.	
Traceability	REQ-POM-FUN -02	

7.4.3 Update a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To update an existing network policy.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is a network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to update a policy.	
Step 1 (M)	The IPRManager makes a decision to update the existing policy.	
Step 2 (M)	After the completion of the policy update process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy is updated and stored in the IRPAgent.	
Traceability	REQ-POM-FUN -03	

7.4.4 Query a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To query an existing network policy.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is a network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to query a policy.	
Step 1 (M)	The IPRManager makes a decision to query the existing policy.	
Step 2 (M)	After the completion of the policy query process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy information is returned to the IPRManager.	
Traceability	REQ-POM-FUN -04	

7.4.5 Activate a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To activate an existing network policy.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is a network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to active a policy.	
Step 1 (M)	The IPRManager makes a decision to active the existing policy.	
Step 2 (M)	After the completion of the policy activation process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy is activated in the IRPAgent.	
Traceability	REQ-POM-FUN -05	

7.4.6 Deactivate a policy in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To query an existing network policy.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is an active network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to deactive a policy.	
Step 1 (M)	The IPRManager makes a decision to deactive the existing policy.	
Step 2 (M)	After the completion of the policy deactivation process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy is deactivated in the IRPAgent.	
Traceability	REQ-POM-FUN -06	

7.4.7 Reporting policy conflicts in the context of NFV

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	To query an existing network policy.	
Actors and Roles	IRPManager as user	
Telecom resources	3GPP management system (NM, EM).	
Assumptions	3GPP management systems are running normally, scenarios and policy requirements are clearly defined.	
Pre conditions	The systems is correctly configured and normally running. There is a network policy stored in the IRPAgent.	
Begins when	The IRPManager decides to update a policy.	
Step 1 (M)	The IPRManager makes a decision to query the existing policy.	
Step 2 (M)	After the completion of the policy conflicts notification process, the IRPAgent informs the IRPManager on the result of the process.	
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	The policy conflicts notification is returned to the IRPManager.	
Traceability	REQ-POM-FUN -07	

8 Policy management procedures

8.1 General description for policy management procedures

There are two kinds of management procedures. The first kind includes those operations that allow the NM to invoke policy management operations towards the EM.

- Policy Creation
- Policy Deletion
- Policy Update
- Policy Query
- Policy Activation

- Policy Deactivation

The second kind includes the operation that the EM notifies NM policy conflicts found in EM.

- Policy Conflicts Notification.

8.2 Policy Creation

After the NM (IRP Manager) decides to create a policy (ies), it will send the create request to the EM (IRP Agent).

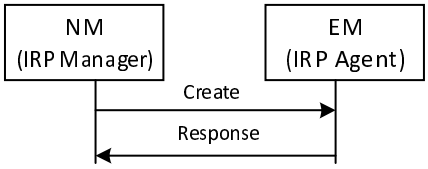


Figure 8.2: policy creation message flow

8.3 Policy Deletion

After the NM (IRP Manager) decides to delete a policy (ies), it will send the delete request to the EM (IRP Agent).

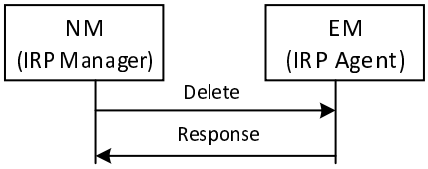


Figure 8.3: policy deletion message flow

8.4 Policy Update

After the NM (IRP Manager) decides to update a policy (ies), it will send the update request to the EM (IRP Agent).

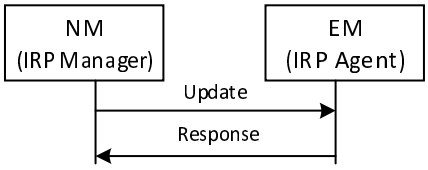


Figure 8.4: policy update message flow

8.5 Policy Query

After the NM (IRP Manager) decides to query a policy, it will send the query request to the EM (IRP Agent). For the query request, the EM (IRP Agent) sends back the response with the policy to the NM (IRP Manager).

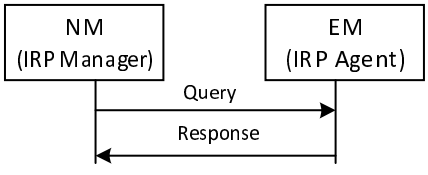


Figure 8.5: policy query message flow

8.6 Policy Activation

After the NM (IRP Manager) decides to activate a policy (ies), it will send the policy activation request to the EM (IRP Agent).

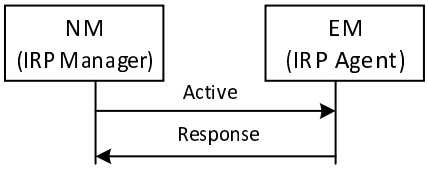


Figure 8.6: policy activation message flow

8.7 Policy Deactivation

After the NM (IRP Manager) decides to deactivate a policy (ies), it will send the policy deactivation request to the EM (IRP Agent).

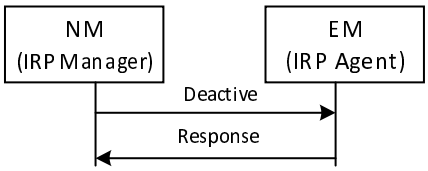


Figure 8.7: policy deactivation message flow

8.8 Policy Conflicts Notification

When the EM (IRP Agent) receives new or updated policy (ies) from NM (IRP Manager), it will check whether the each received policy is conflicts with the previous storage policy. If the conflict is detected, the EM (IRP Agent) will notify the information to the NM (IRP Manager).

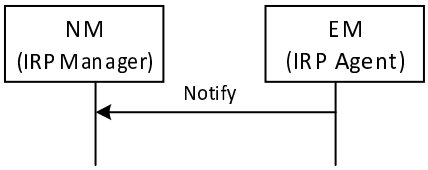


Figure 8.8: policy conflicts notification message flow

9 Information Object Classes

9.1 Information entities imported and local label

Label reference	Local label
TS 32.312 [3], information object class, `ManagedGenericIRP`	ManagedGenericIRP

9.2 Class diagram

9.2.1 Attributes and relationships

This clause introduces the set of Information Object Classes (IOCs) that encapsulate capabilities contained within the IRP. This clause provides the overview of all support object classes in UML. Subsequent clauses provide more detailed specification of various aspects of these support object classes.

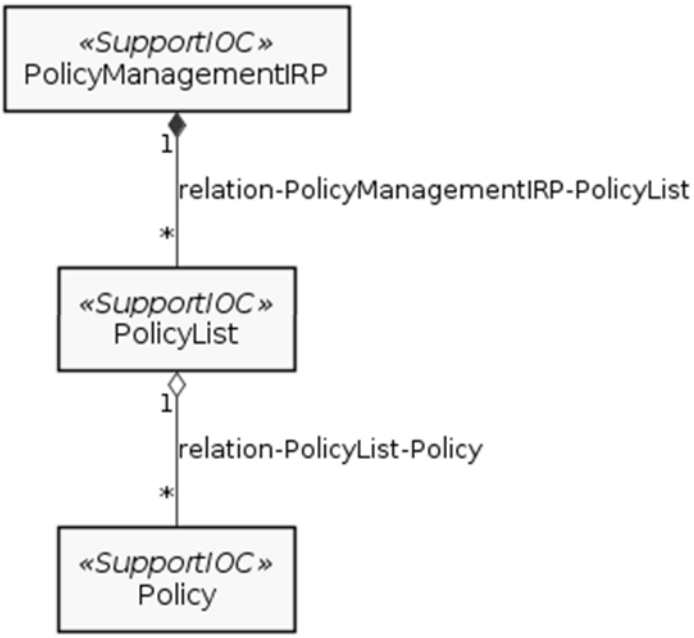


Figure 9.2.1: Information Object Class UML Diagram

9.2.2 Inheritance

This IOC inherits from ManagedGenericIRP IOC specified in TS.32.312 [3].

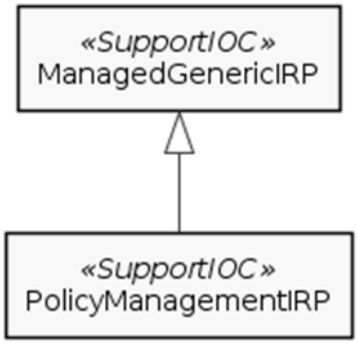


Figure 9.2.2: Information Object Class UML Diagram

9.3 Information object classes definition

9.3.1 Policy

9.3.1.1 Definition

This IOC represents a network policy.

9.3.1.2 Attributes

Attribute name	Support Qualifier	isReadble	isWriteble	isInvariant	isNotifyable
policyId	M	T	F	T	T
policyPriority	M	T	T	F	T
policyStatus	M	T	T	F	T
policyType	M	T	T	F	T
policyContent	M	T	T	F	T

9.3.2 PolicyList

9.3.2.1 Definition

PolicyList is the representation of the list of Policy.

9.3.2.2 Attributes

Attribute name	Support Qualifier	isReadble	isWriteble	isInvariant	isNotifyable
policyListId	M	T	F	T	T
policyIdList	M	T	F	T	T

9.3.3 PolicyManagementIRP

9.3.3.1 Definition

PolicyManagementIRP is the representation of the policy management capabilities specified by the present document. This Support IOC inherits from ManagedGenericIRP Support IOC specified in 3GPP TS 32.312 [x].

9.3.3.2 Attribute

There is no additional attribute defined for this IOC besides those inherited.

9.3.3.3 Notifications

Name	Qualifier	Notes
policyConflictNotification	M	If there is network policy conflict(s), then the IRPAgent may send this notification to the IPRmanager

9.4 Information relationship definitions

9.4.1 relation-PolicyManagementIRP-PolicyList (M)

9.4.1.1 Definition

This represents the relationship between PolicyManagementIRP and PolicyList.

9.4.1.2 Role

Name	Definition
thePolicyList	It represents the PolicyList.
thePolicyManagementIRP	It represents the PolicyManagementIRP.

9.4.1.3 Constraint

Name	Definition
uniquePolicyListId	The policyListIds playing the role of thePolicyList are unique within a particular PolicyManagementIRP.

9.4.2 relation-PolicyList-Policy (M)

9.4.2.1 Definition

This represents the relationship between PolicyList and Policy.

9.4.2.2 Role

Name	Definition
thePolicy	It represents the Policy.
thePolicyList	It represents the PolicyList.

9.4.2.3 Constraint

Name	Definition
uniquePolicyId	The policyIds of all Policys, playing the role of the Policy, are unique within a particular PolicyManagementIRP and PolicyList.

9.5 Information attribute definitions

9.5.0 Introduction

This clause defines the semantics of the attributes used in SupportIOCs.

9.5.1 Definitions and legal values

Attribute Name	Definition	Legal Values
policyId	It identifies the Policy instance (and distinguishes it from all other existing and stopped Policy instances of the PolicyManagementIRP Agent).	Any identifier. Value type is string.
policyPriority	It specifies the priority of Policy	Its value should be one of the following: Low, Medium, High
policyStatus	It specifies the status of Policy. If a policy is activated, and then its status is active. If it is deactivated ,then its status is deactivated	Its value should be one of the following: Activated, Deactivated
policyType	It identifies a name of one policy type	Its value is network operator specific. Value type is string.
policyContent	It identifies the content of a network policy	Its value is network operator specific. Value type is string.
policyListId	It identifies the singleton PolicyList of the PolicyManagementIRP Agent.	Any identifier. Value type is string.
policyIdList	It is an array list of Policy.policyId.	Value type is array.
policyConflictNotification	It specifies which policies conflict.	Its value is vendor specific. Value type is string.

10 Interface definition

10.1 Class diagram

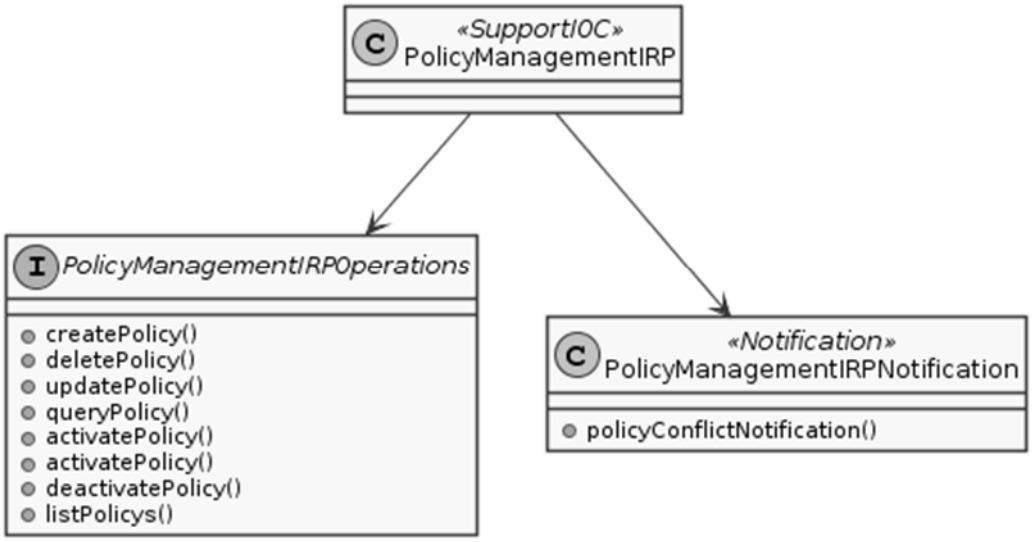


Figure 10.1: Class Diagram

10.2 PolicyManagementIRPOperations_Interface (M)

10.2.1 Operation createPolicy (M)

10.2.1.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.2.1.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
policyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values).
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.2.1.3 Results

In case of success, the corresponding policy information is created by the EM (IRP Agent). In case of failure, appropriate error information is returned.

10.2.2 Operation deletePolicy (M)

10.2.2.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
policyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values).

10.2.2.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
policyId	M	Policy.policyId	The policyId of the deleted policy. See clause 9.5.1 (definitions and legal values).
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.2.2.3 Results

In case of success, the policy information are deleted by the EM (IRP Agent), and the corresponding deleted policy information is returned to the NM (IRP Manager). In case of failure, appropriate error information is returned.

10.2.3 Operation updatePolicy (M)

10.2.3.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	O	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	O	Policy.name	See clause 9.5.1 (definitions and legal values).
policyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values).This parameter's value is not allowed to be updated.
policyPriority	O	Policy.policyPriority	See clause 9.5.1 (definitions and legal values). This parameter's value is allowed to be updated.
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values). This parameter's value is allowed to be updated.
policyType	O	Policy.policyType	See clause 9.5.1 (definitions and legal values). This parameter's value is allowed to be updated.
policyContent	O	Policy.policyContent	See clause 9.5.1 (definitions and legal values). This parameter's value is allowed to be updated.

10.2.2.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
policyId	M	Policy.policyId	The policyId of the deleted policy. See clause 9.5.1 (definitions and legal values).
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.2.2.3 Results

In case of success, the corresponding policy information is updated by the EM (IRP Agent). In case of failure, appropriate error information is returned.

10.2.4 Operation queryPolicy (M)

10.2.4.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
policyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values).

10.2.4.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
policyId	M	Policy.policyId	The policyId of the deleted policy. See clause 9.5.1 (definitions and legal values).
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.2.4.3 Results

In case of success, the corresponding policy information is returned by the EM (IRP Agent). In case of failure, appropriate error information is returned.

10.2.5 Operation activatePolicy (M)

10.2.5.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
policyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values).

10.3.5.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
PolicyId	M	Policy.policyId	See clause 9.5.1 (definitions and legal values). Identifier of the activated Policy.
policyPriority	M	Policy.policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy.policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy.policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy.policyContent	See clause 9.5.1 (definitions and legal values).

10.3.5.3 Results

In case of success, the policy(ies) information are activated by the EM (IRP Agent) , and the activated policy information is returned to the NM (IRP Manager). In case of failure, appropriate error information is returned.

10.2.6 Operation deactivatePolicy (M)

10.2.6.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
policyId	M	Policy. policyId	See clause 9.5.1 (definitions and legal values).

10.2.6.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
designer	M	Policy.designer	See clause 9.5.1 (definitions and legal values).
name	M	Policy.name	See clause 9.5.1 (definitions and legal values).
PolicyId	M	Policy. policyId	See clause 9.5.1 (definitions and legal values). Identifier of the activated Policy.
policyPriority	M	Policy. policyPriority	See clause 9.5.1 (definitions and legal values).
policyStatus	M	Policy. policyStatus	See clause 9.5.1 (definitions and legal values).
policyType	M	Policy. policyType	See clause 9.5.1 (definitions and legal values).
policyContent	M	Policy. policyContent	See clause 9.5.1 (definitions and legal values).

10.2.6.3 Results

In case of success, the policy(ies) information are deactivated by the EM (IRP Agent), and the deactivated policy information is returned to the NM (IRP Manager). In case of failure, appropriate error information is returned.

10.2.7 Operation queryPolicyList (M)

10.2.7.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
policyListId	M	PolicyList.policyListId	See clause 9.5.1 (definitions and legal values).

10.2.7.2 Output parameters

Parameter Name	Qualifier	Matching Information	Comment
policyListId	M	PolicyList.policyListId	See clause 9.5.1 (definitions and legal values).
policyIdList	M	PolicyList.policyIdList	See clause 9.5.1 (definitions and legal values).

10.2.7.3 Results

In case of success, the EM (IRP Agent) returns the corresponding policy information. In case of failure, appropriate error information is returned.

10.3 PolicyManagementIRPNotification_Interface (M)

10.3.1 policyConflictNotification

10.3.1.1 Input parameters

Parameter Name	Qualifier	Matching Information	Comment
activatedPolicyList	M	List of <PolicyList.policyListId> and their corresponding policy is activated.	See clause 9.5.1 (definitions and legal values). Identifier(s) of the activated Policy.

10.3.1.2 Triggering Event

10.3.1.2.1 From-state

internalProblem

Assertion Name	Definition
internalProblem	Because of a network policy conflict's problem, PolicyManagementIRP Agent decides that it no longer can maintain the policy in any policyStatus but "deactivated".

10.3.1.2.2 To-state

policyConflictNotification

Assertion Name	Definition
policyConflictNotification	There is network policyconflict(s).

11 Solution Set (SS) definitions

The present document defines the following policy management Solution Set Definitions:

- Annex A provides the XML Definitions.
- Annex B provides the SOAP Solution Set.

Annex A (normative): XML definitions

A.1 General

This annex contains the XML definitions for the policy management Integration Reference Point.

A.2 Architectural features

The overall architectural feature of policy management IRP is specified, this clause specifies features that are specific to the Schema definitions.

The present document provides the main part of the XML file format definition for the policy management IRP IS.

The other parts of this XML definition are NRM-specific parts. All NRM IRPs that include SS-level XML definition are in the scope of the policy management IRP.

A.3 Mapping

A.3.1 General mapping

An IOC maps to an XML element of the same name as the IOC's name in the Information Model. An IOC attribute maps to a sub-element of the corresponding IOC's XML element, and the name of this sub-element is the same as the attribute's name in the Information Model.

A.3.2 Information Object Class (IOC) mapping

The mapping is not present in the current version of the present document.

A.4 Solution Set definitions

A.4.1 XML definition structure

Clause A.4.2 provides a graphical representation of the XML elements.

Clause A.4.3 provides the schema fragment for policy management (policy creation/deletion/update/query/activation/deactivation/conflicts notification) XML files.

A.4.2 Graphical Representation

The graphical representation is not present in the current version of the present document.

A.4.3 XML Schema “PolicyManagement.xsd”

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
    3GPP TS 28.311 policy management XML Schema
    PolicyOperation.xsd
-->
```

```
<schema xmlns:xb="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#PolicyManagement"
xmlns:xe="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#PolicyManagement "
xmlns="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#PolicyManagement"
elementFormDefault="qualified" attributeFormDefault="unqualified">

        <!—createPolicy Request -->
        <element name="createPolicy">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- createPolicy Response -->
        <element name=" createPolicyResponse">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- deletePolicy Request -->
        <element name="deletePolicy">
            <complexType>
                <sequence>
                    <element name="policyId" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- deletePolicy Response -->
        <element name="deletePolicyResponse">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- updatePolicy Request -->
        <element name="updatePolicy">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- updatePolicy Response -->
        <element name="updatePolicy">
            <complexType>
                <sequence>
```

```
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- queryPolicy Request -->
<element name="queryPolicy">
    <complexType>
        <sequence>
            <element name="policyId " type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- queryPolicy Response -->
<element name="queryPolicyResponse">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- activatePolicy Request -->
<element name="activatePolicy">
    <complexType>
        <sequence>
            <element name="policyId " type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- activatePolicy Response -->
<element name=" activatePolicyResponse">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- deactivatePolicy Request -->
<element name="deactivatePolicy">
    <complexType>
        <sequence>
            <element name="policyId" type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- deactivatePolicy Response -->
<element name=" deactivatePolicyResponse">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
```

```
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- queryPolicyList Request -->
        <element name="queryPolicyList">
            <complexType>
                <sequence>
                    <element name="policyListId" type="Identifier"/>
                </sequence>
            </complexType>
        </element>

        <!-- queryPolicyList Response -->
        <element name="queryPolicyList">
            <complexType>
                <sequence>
                    <element name="policyListId" type="Identifier"/>
                    <element name="policyIdList" type="array"/>
                </sequence>
            </complexType>
        </element>

        <!-- policyConflictNotification Request -->
        <element name="policyConflictNotification">
            <complexType>
                <sequence>
                    <element name="activatedPolicyList" type="Identifier"/>
                </sequence>
            </complexType>
        </element>

</schema>
```

Annex B (normative): SOAP Solution Set

B.1 General

This annex contains the XML definitions for the policy management Integration Reference Point.

B.2 Architectural features

The overall architectural feature of the policy management IRP is specified in the present document. This clause specifies features that are specific to the SOAP Solution Set.

The present document uses "document" style in the WSDL description.

The present document uses "literal" encoding style in the WSDL description.

B.3 Mapping

B.3.1 Operation and notification mapping

Policy management IRP management: Clause 9 defines semantics of operation visible across the Itf-N. Table B.3 indicates mapping of these operations to their equivalents defined in this SS.

Table B.3: Mapping from IS Operation to SS Equivalents

IS Operation	SS Operation	Qualifier
createPolicy	createPolicy	M
deletePolicy	deletePolicy	M
updatePolicy	updatePolicy	M
queryPolicy	queryPolicy	M
activatePolicy	activatePolicy	M
deactivatePolicy	deactivatePolicy	M
queryPolicyList	queryPolicyList	M
policyConflictNotification	policyConflictNotification	M

B.3.2 Operation parameter mapping

B.3.2.0 General

The clause 10 defines semantics of parameters carried in operations. The tables below show the mapping of these parameters, as per operation, to their equivalents defined in this SS.

B.3.2.1 Operation `createPolicy`

B.3.2.1.1 Input parameters

Table B.3.2.1.1: Mapping from IS createPolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.1.2 Output parameters

Table B.3.2.1.2: Mapping from IS createPolicy output parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.2 Operation `deletePolicy`

B.3.2.2.1 Input parameters

Table B.3.2.2.1: Mapping from IS deletePolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyId	policyInfoId	M

B.3.2.2.2 Output parameters

Table B.3.2.2.2: Mapping from IS deletePolicy output parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.3 Operation updatePolicy

B.3.2.3.1 Input parameters

Table B.3.2.3.1: Mapping from IS updatePolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	O
name	name	O
policyId	policyInfoId	M
policyPriority	policyPriority	O
policyStatus	policyStatus	M
policyType	policyType	O
policyContent	policyContent	O

B.3.2.3.2 Output parameters

Table B.3.2.3.2: Mapping from IS updatePolicy output parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.4 Operation queryPolicy

B.3.2.4.1 Input parameters

Table B.3.2.4.1: Mapping from IS queryPolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyId	policyInfoId	M

B.3.2.4.2 Output parameters

Table B.3.2.4.2: Mapping from IS queryPolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.5 Operation activatePolicy

B.3.2.5.1 Input parameters

Table B.3.2.5.1: Mapping from IS activatePolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyId	policyInfoId	M

B.3.2.5.2 Output parameters

Table B.3.2.5.2: Mapping from IS startSession input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.6 Operation deactivatePolicy

B.3.2.6.1 Input parameters

Table B.3.2.6.1: Mapping from IS deactivatePolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyId	policyInfoId	M

B.3.2.6.2 Output parameters

Table B.3.2.6.2: Mapping from IS deactivatePolicy input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
designer	designer	M
name	name	M
policyId	policyInfoId	M
policyPriority	policyPriority	M
policyStatus	policyStatus	M
policyType	policyType	M
policyContent	policyContent	M

B.3.2.7 Operation queryPolicyList

B.3.2.7.1 Input parameters

Table B.3.2.7.1: Mapping from IS listPolicys input parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyListId	policyListId	M

B.3.2.7.2 Output parameters

Table B.3.2.7.2: Mapping from IS listPolicys output parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
policyListId	policyListId	M
policyIdList	policyIdList	M

B.3.2.8 Operation policyConflictNotification

B.3.2.8.1 Input parameters

Table B.3.2.8.1: Mapping from IS policyConflictNOtificaition output parameters to SS equivalents

IS Operation parameter	SS Operation parameter	Qualifier
activatedPolicyList	activatedPolicyList	M

B.3.2.8.2 Output parameters

Void.

B.4 Solution Set definitions

B.4.1 WSDL definition structure

The present document defines the main part of what are supported by the policy management IRP agent.

B.4.2 Graphical Representation

The graphical representation is not present in the current version of the present document.

B.4.3 WSDL specification “PolicyManagementIRPSystem.wsdl”

```
<?xml version="1.0" encoding="UTF-8"?>
<definitions xmlns="http://schemas.xmlsoap.org/wsdl/"
xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:PolicyManagement
IRPSystem=http://www.3gpp.org/ftp/Specs/archive/28_series/28.311/schema/28311/PolicyManagementIRPSys
tem
xmlns:xs="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://www.3gpp.org/ftp/Specs/archive/28_series/28.311/schema/28311/GenericIRPSyste
m">
   <types>
        <!—createPolicy Request -->
        <element name="createPolicy">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- createPolicy Response -->
        <element name=" createPolicyResponse">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- deletePolicy Request -->
        <element name="deletePolicy">
            <complexType>
                <sequence>
                    <element name="policyId" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- deletePolicy Response -->
        <element name="deletePolicyResponse">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- updatePolicy Request -->
        <element name="updatePolicy">
            <complexType>
                <sequence>
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
```

```
</element>

<!-- updatePolicy Response -->
<element name="updatePolicy">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- queryPolicy Request -->
<element name="queryPolicy">
    <complexType>
        <sequence>
            <element name="policyId " type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- queryPolicy Response -->
<element name="queryPolicyResponse">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- activatePolicy Request -->
<element name="activatePolicy">
    <complexType>
        <sequence>
            <element name="policyId " type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- activatePolicy Response -->
<element name=" activatePolicyResponse">
    <complexType>
        <sequence>
            <element name="designer" type="string"/>
            <element name="name" type="string"/>
            <element name="policyId " type="Identifier"/>
            <element name="policyPriority" type="string"/>
            <element name="policyStatus" type="string"/>
            <element name="policyType" type="string"/>
            <element name="policyContent" type="string"/>
        </sequence>
    </complexType>
</element>

<!-- deactivatePolicy Request -->
<element name="deactivatePolicy">
    <complexType>
        <sequence>
            <element name="policyId " type="Identifier"/>
        </sequence>
    </complexType>
</element>

<!-- deactivatePolicy Response -->
<element name=" deactivatePolicyResponse">
    <complexType>
        <sequence>
```

```
                    <element name="designer" type="string"/>
                    <element name="name" type="string"/>
                    <element name="policyId " type="Identifier"/>
                    <element name="policyPriority" type="string"/>
                    <element name="policyStatus" type="string"/>
                    <element name="policyType" type="string"/>
                    <element name="policyContent" type="string"/>
                </sequence>
            </complexType>
        </element>

        <!-- queryPolicyList Request -->
        <element name="queryPolicyList">
            <complexType>
                <sequence>
                    <element name="policyListId" type="Identifier"/>
                </sequence>
            </complexType>
        </element>

        <!-- queryPolicyList Response -->
        <element name="queryPolicyList">
            <complexType>
                <sequence>
                    <element name="policyListId" type="Identifier"/>
                    <element name="policyIdList" type="array"/>
                </sequence>
            </complexType>
        </element>

        <!-- policyConflictNotification Request -->
        <element name="policyConflictNotification">
            <complexType>
                <sequence>
                    <element name="activatedPolicyList" type="Identifier"/>
                </sequence>
            </complexType>
        </element>

    </types>

    <message name="createPolicy Request">
       <part name="parameter" element="createPolicy"/>
    </message>
    <message name="createPolicy Response">
       <part name="parameter" element="createPolicyResponse"/>
    </message>
    <message name="deletePolicy Request ">
       <part name="parameter" element="deletePolicy"/>
    </message>
    <message name="deletePolicy Response">
       <part name="parameter" element="deletePolicyResponse"/>
    </message>
    <message name="updatePolicy Request ">
       <part name="parameter" element="PolicyUpdate"/>
    </message>
    <message name="updatePolicy Response">
       <part name="parameter" element="updatePolicyResponse"/>
    </message>
    <message name="queryPolicyRequest ">
       <part name="parameter" element="queryPolicy"/>
    </message>
    <message name="queryPolicyResponse">
       <part name="parameter" element="queryPolicyResponse"/>
    </message>
    <message name="activePolicyRequest ">
       <part name="parameter" element="ActivePolicy"/>
    </message>
    <message name="activePolicyResponse">
       <part name="parameter" element="ActivePolicyResponse"/>
    </message>
    <message name="deactivePolicyRequest ">
       <part name="parameter" element="deactivePolicy"/>
    </message>
    <message name="deactivePolicyResponse">
       <part name="parameter" element="deactivePolicyResponse"/>
    </message>
    <message name="queryPolicyListRequest">
```

```
      <part name="parameter" element="queryPolicyList"/>
   </message>
   <message name="queryPolicyListResponse">
      <part name="parameter" element="queryPolicyListResponse"/>
   </message>
   <message name="policyConflictNotificaitonRequest">
      <part name="parameter" element="policyConflictsNotificaiton"/>
   </message>
   <portType name="PolicyMangementIRPPortType">
      <operation name="createPolicy">
         <input message="PolicyMangementIRPSystem:createPolicyRequest"/>
         <output message="PolicyMangementIRPSystem:createPolicyResponse"/>
      </operation>
      <operation name="deletePolicy">
         <input message="PolicyMangementIRPSystem:deletePolicyRequest"/>
         <output message="PolicyMangementIRPSystem:deletePolicyResponse"/>
      </operation>

      <operation name="updatePolicy">
         <input message="PolicyMangementIRPSystem:updatePolicyRequest"/>
         <output message="PolicyMangementIRPSystem:updatePolicyResponse"/>
      </operation>
      <operation name="queryPolicy">
         <input message="PolicyQueryIRPSystem:queryPolicyRequest"/>
         <output message="PolicyQueryIRPSystem:queryPolicyResponse"/>
      </operation>
      <operation name="activePolicy">
         <input message="PolicyMangementIRPSystem:activePolicyRequest"/>
         <output message="PolicyMangementIRPSystem:activePolicyResponse"/>
      </operation>
      <operation name="deactivePolicy">
         <input message="PolicyMangementIRPSystem:deactivePolicyRequest"/>
         <output message="PolicyMangementIRPSystem:deactivePolicyResponse"/>
      </operation>
      <operation name="queryPolicyList">
         <input message="PolicyMangementIRPSystem:queryPolicyListRequest"/>
         <output message="PolicyMangementIRPSystem:queryPolicyListResponse"/>
      </operation>

      <operation name="PolicyConflictNotification">
         <input message="PolicyMangementIRPSystem:policyConflictNotificationRequest"/>
      </operation>
   </portType>
   <binding name="PolicyManagementIRPBinding" type="PolicyManagementIRPSystem:
PolicyManagementIRPPortType">
        <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http"/>
        <operation name="createPolicy">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
createPolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="deletePolicy">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
deletePolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="updatePolicy">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
updatePolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="queryPolicy">
```

```
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
queryPolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="activePolicy">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
activePolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="deactivePolicy">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
deactivePolicy" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
        <operation name="queryPolicyList">
            <soap:operation soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#
queryPolicyList" style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>

        <operation name="policyConflictNofitication">
            <soap:operation
soapAction="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#PolicyConflictNofitication"
style="document"/>
            <input>
                <soap:body use="literal"/>
            </input>
            <output>
                <soap:body use="literal"/>
            </output>
        </operation>
    </binding>
    <service name="PolicyMangementIRPService">
         <port name="PolicyMangementIRPPort" binding="PolicyMangementIRPSystem:
    PolicyMangementIRPBinding">
            <soap:address
    location="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#BulkCMIRP"/>
         </port>
         <port name="GenericIRPPort" binding="genericIRPSystem:GenericIRPBinding">
            <soap:address
    location="http://www.3gpp.org/ftp/specs/archive/28_series/28.311#GenericIRP"/>
         </port>
    </service>
</definitions>
```

Annex C (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2018-09						Skeleton	0.0.0
2018-10	SA5#121	S5-186398 S5-186468				Update to implement the agreed pCRs in SA5#121: S5-186398 pCR 28.311 Add Scope S5-186468 Rel-16 draft TS Skeleton 28.311	0.1.0
2018-11	SA5#122	S5-187385 S5-187386				Update to implement the agreed pCRs in SA5#122: S5-187385 Add Policy management architecture S5-187386 Add Business level requirements	0.2.0
2019-06	SA5#125 Adhoc	S5-194428 S5-194429 S5-194430 S5-194431 S5-194432 S5-194433 S5-194434 S5-194435				Update to implement the agreed pCRs in SA5#125Adhoc: S5-194428 Policy Management Procedures S5-194429 Policy Creation S5-194430 Policy Deletion S5-194431 Policy Update S5-194432 Policy Query S5-194433 Policy Activation S5-194434 Policy Deactivation S5-194435 Policy Conflicts Notification	0.3.0
2019-08	SA5#126	S5-195179 S5-195843 S5-195845 S5-195846 S5-195847				Update to implement the agreed pCRs in SA5#126: S5-195179 Add abbreviations S5-195843 Add references S5-195845 Add overview S5-195846 Rewrite business level requirements S5-195847 Add specification level requirements	0.4.0
2019-10	SA5#127	S5-196658				Update to implement the agreed pCRs in SA5#127: S5-196658 Add Solution Set(SS) definitions	0.5.0
2019-11	SA5#128	S5-197763 S5-197764 S5-197765 S5-197766				Update to implement the agreed pCRs in SA5#128: S5-197763 pCR 28.311 Add Information Object Classes S5-197764 pCR 28.311 Add XML definition S5-197765 pCR 28.311 Add SOAP Solution Set S5-197766 pCR 28.311 Rewrite interface	0.6.0
2019-12	SA#86	SP-191185				Presented for information and approval	1.0.0
2019-12	SA#86					Change control version	16.0.0

History

Document history		
V16.0.0	November 2020	Publication